



ALKALINE HYDROLYSIS OF POLYETHYLENE TEREPHTHALATE UNDER MICROWAVE IRRADIATION

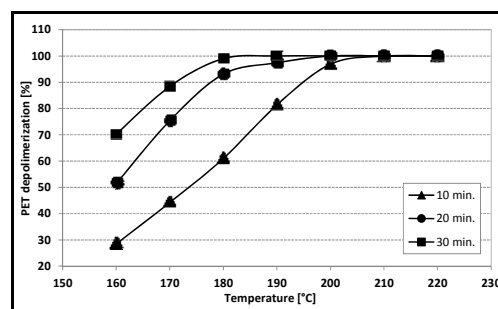
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Received October 31, 2016

In this work we focus on the high-yield recovery of terephthalic acid through alkaline hydrolysis of PET waste using microwave irradiation. The main product is TPA disodium salt, which is converted to TPA by treatment with acid. The influence of reaction time, temperature and the amount of alkaline solution on the TPA yield have been determined in order to establish the optimum parameters at which the reaction is completed. PET depolymerization was carried out under microwave irradiation at 160–220°C, reaction time ranging from 2 to 120 minutes. The obtained TPA was analyzed using FT-IR spectroscopy, X-ray diffraction, ¹H NMR, and ¹³C NMR. Percent depolymerization of PET and recovery yields of TPA were gravimetric determined. The use of microwave irradiation for PET depolymerization by alkaline hydrolysis provides the significant reduction in reaction time (less than 30 minutes) as compared with the conventional heating methods. Depolymerization of PET waste was complete and TPA yield reached more than 98%.



INTRODUCTION

Polyethylene terephthalate (PET) is thermoplastic polyester non-degradable in normal conditions. The overall world production of polyesters was 25–30 million tons in 2000, this value increased to 55 million tons in 2012 and mostly consisted of polyethylene terephthalate.¹ PET is primarily used as fibers, sheets and films, food and beverage packaging, electronics, automotive parts, houseware, lighting products, power tools, sports goods, photographic applications, X-ray sheets and textiles.^{2,3} Polyethylene terephthalate has become the most advantageous packaging material worldwide for water and softdrinks bottles due to its excellent properties such as unbreakability, high clarity, very low weight, as well as good barrier properties

towards moisture and oxygen compared to other packaging materials.⁴ Although PET is a non-toxic material, its non-biodegradability and the large amount of waste is a serious problem concerning environmental pollution. Therefore, recycling processes are the best option to economically reduce the amount of PET waste.

Among various recycling methods, chemical recycling involves transformation of polymer chain using different reagents including water, alcohols, methanol, etc. leading to total depolymerization of PET to the monomers, or partial depolymerization to oligomers and other chemical products.⁵ There is a growing interest in hydrolysis for the chemical recycling of PET, since it is the only method that leads to the monomers from which PET is produced, terephthalic acid (TPA) and ethylene

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glycol (EG). The main disadvantage of this method is the use of high temperatures (200–250°C) and pressures (1.4–2 MPa) as well as the long time needed for complete depolymerization.⁶

Heating by microwave irradiation as compared with conventional heating provides some advantages such as rapid heating with high specificity without contact with material, shorter reaction times, great convenience and reduced costs.⁷ Microwave irradiation has been used for glycolytic depolymerization of PET with ethylene glycol⁸ or diethylene glycol,⁹ aminolytic depolymerization using excess of ethanolamine,¹⁰ methanolysis with methanol and zinc acetate catalyst,¹¹ depolymerization in neutral water^{12,13} or alkaline solution.¹⁴

In our previous work,¹⁵ hydrothermal depolymerization of PET waste in sub- and supercritical conditions was studied. In the present study, hydrolytic depolymerization of PET waste in alkaline solution using microwave irradiation as heating technique was investigated in order to obtain pure terephthalic acid in high yields.

EXPERIMENTAL

Colorless post-consumer PET bottles were manually cut into small flakes (1÷3 mm × 1÷3 mm), then washed and dried. 1g PET waste flakes and various amounts of aqueous solution of NaOH (5–10 mL) were charged in high-pressure teflon reactors of 60 mL capacity without stirring. PET depolymerization was carried out under microwave irradiation using a microwave digestion system speedwave MWS – 2 (Berghof, Germany). The following program was used: step 1 (preheating): 160°C, P = 80%, 5 min.; step 2: 160–220°C, P = 90% and time in the range 2–120 min.

The reaction mixture was filtered to separate unreacted PET. Residual PET was washed with distilled water and dried.

Percent depolymerization of PET was determined by gravimetry and defined as:

$$\text{PET}_{\text{depolymerization}} (\%) = (W_{\text{PET}}^0 - W_{\text{PET}}^t) / W_{\text{PET}}^0 \times 100 \quad (1)$$

where W_{PET}^0 is the initial weight of PET and W_{PET}^t is the weight of PET remaining after reaction time t .

The filtrate was acidified with concentrated HCl in order to precipitate TPA. The white precipitate of TPA was filtered, washed with double-distilled water, dried and weighed. The recovery yield of TPA was calculated according to the following formula:

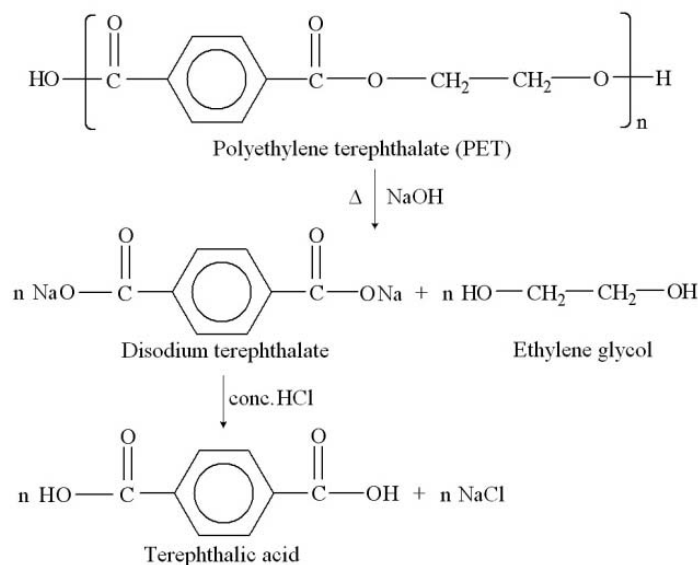
$$\eta_{\text{TPA}} (\%) = W_{\text{TPA}}^{\text{exp}} / W_{\text{TPA}}^t \times 100 \quad (2)$$

where $W_{\text{TPA}}^{\text{exp}}$ is the weight of TPA obtained experimentally and W_{TPA}^t is the theoretical weight of TPA that could be obtained for complete depolymerization of PET.

The reaction products were analyzed by using FT-IR spectroscopy (VERTEX 70, Bruker), X-ray diffraction (X'Pert PRO MPD, PANalytical) and scanning electron microscopy (SEM) (Inspect S PANalytical). Also, for verification of chemical structure of the final product, ¹³C-NMR and ¹H-NMR spectra were recorded on a Bruker Avance DRX 400 device with the operating frequencies of 400.13 (for ¹H) and 100.61 (for ¹³C) by using as solvents CDCl₃ for ¹³C-NMR and DMSO-d₆ for ¹H-NMR. The results were compared with those obtained for authentic TPA.

RESULTS AND DISCUSSION

Waste PET flakes were hydrolyzed in sodium hydroxide to obtain the disodium terephthalate salt (TPA-Na₂) and ethylene glycol (EG), according to the chemical reaction shown in Scheme 1. The TPA-Na₂ salt was then acidified with concentrated HCl in order to precipitate TPA.



Scheme 1 – Hydrolytic depolymerization reaction of PET in alkaline solution.

Effect of the reaction parameters

A series of PET degradation experiments under microwave irradiation without catalyst at autogenous pressure were performed in order to establish the effect of reaction time, temperature and the amount of NaOH solution on the degree of depolymerization. The amounts of residual PET and TPA obtained after purification were determined gravimetrically. However, it was difficult to accurately measure the amount of solid product obtained in each case, because of inevitable weight losses during the product processing. The first tests were conducted at 220°C with 5 mL and 10 mL NaOH solution for 120 minutes, then gradually decreasing the reaction time. The results are presented in Table 1.

The depolymerization of PET flakes was complete in the range 120-5 minutes when it was used 10 mL alkaline solution for hydrolysis

reaction. PET degradation was almost complete (99.1%) for 5 minutes reaction time with 5 mL NaOH solution. Carrying out the reaction for 2 minutes, led to very good PET depolymerization degree, *i.e.* 96.8% when using 10 mL alkaline solution and 95.6% when using 5 mL alkaline solution. The recovery yields of TPA were >97% in both cases. For the experimental tests where PET depolymerization was complete, TPA yield reached more than 98%.

Another set of experimental tests in microwave field aimed to study the influence of temperature on the depolymerization degree of PET waste. The hydrolysis reaction was carried out with 10 mL NaOH solution at different temperatures in the range 160-220°C and reaction times ranging between 10 and 30 minutes. The effect of temperature on the depolymerization degree of PET at different reaction times is shown in Figure 1.

Table 1

Experimental data for PET depolymerization in microwave field at 220°C

No	V _{NaOH 10%} [mL]	t [min]	W _{TPA} ^{exp} [g]	W _{PET} ^t [g]	η _{TPA} [%]
1	10	120	0.8497	-	98.28
2	5	120	0.8512	-	98.46
3	10	60	0.8492	-	98.22
4	5	60	0.8493	-	98.23
5	10	30	0.8489	-	98.19
6	5	30	0.8473	-	98.00
7	10	20	0.8507	-	98.40
8	5	20	0.8481	-	98.09
9	10	10	0.8498	-	98.29
10	5	10	0.8502	-	98.34
11	10	5	0.8515	-	98.49
12	5	5	0.8470	0.0009	97.97
13	10	2	0.8403	0.0325	97.19
14	5	2	0.8382	0.0437	96.95

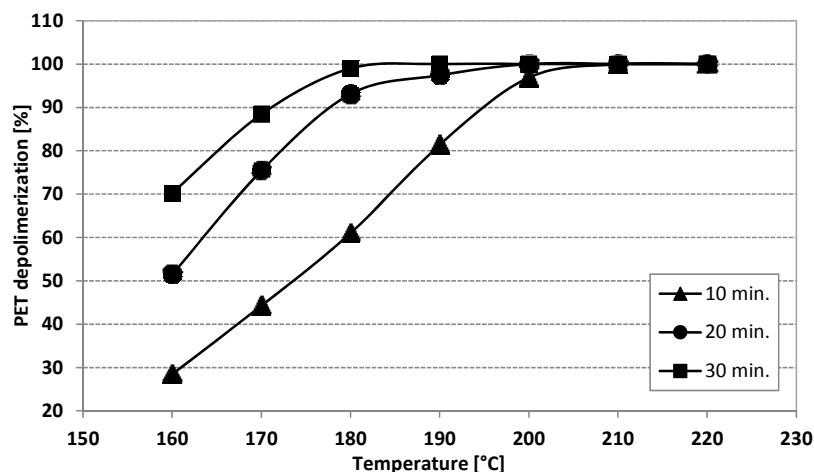


Fig. 1 – Effect of temperature and reaction time on the conversion degree of PET waste.

As expected, a decrease of the reaction temperature leads to a decrease in PET conversion. Increasing reaction time leads to a better decomposition of PET waste. The time required for complete PET depolymerization was 10 minutes at 220°C, 20 minutes at 210 and 200°C, and 30 minutes at 190°C. PET waste depolymerization was almost complete at 210°C after 10 minutes (99.95%). In these cases, TPA yield reached more than 98%. Between 160 and 180°C was not achieved complete degradation of PET waste. The lowest decomposition degree was obtained at 160°C after 10 minutes of reaction (28.4%).

According to the results of this study, the lowest reaction times required for complete PET depolymerization was 5 minute at 220°C and 30 minutes at 190°C. The use of microwave irradiation for PET waste recycling by alkaline

hydrolysis provides the significant reduction in reaction time as compared with the conventional heating methods.

Characterization of the degradation products

The white solid product obtained through PET alkaline hydrolysis after unreacted PET was separated, was characterized by recording of FT-IR spectra in order to be confirmed as terephthalic acid. The FT-IR spectra of obtained product have shown the same characteristic bands as well as standard TPA (Figure 2, Table 2).

The X-ray spectra (Fig. 3) for experimental samples also confirmed the crystalline phase of terephthalic acid for the experimental products.

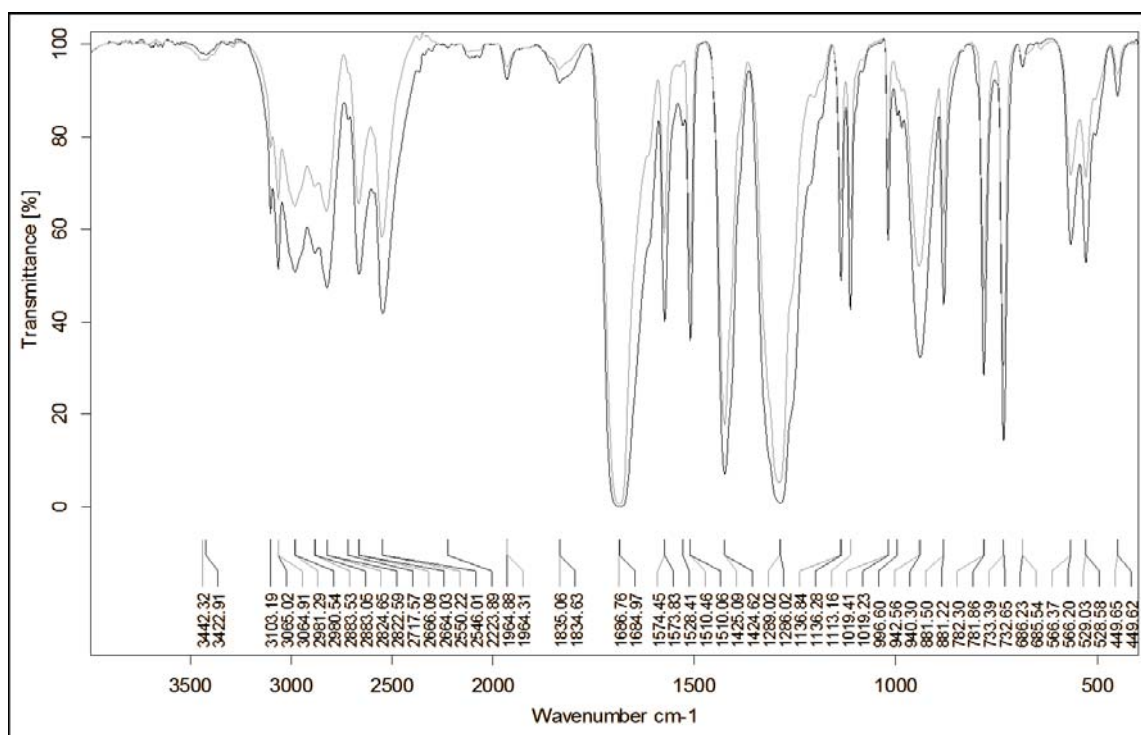


Fig. 2 – FT-IR spectrum of standard TPA (—) and TPA obtained from hydrolytic depolymerization reaction of PET waste (---).

Table 2

Characteristic vibrations of TPA

Vibrations	Assignment
~3065 cm ⁻¹	ν_{C-H} arom
~ 1685 cm ⁻¹	$\nu_{C=O}$
~ 1575, 1510, 1110 cm ⁻¹	1,4 substitution of benzene ring
~ 1285 cm ⁻¹	ν_{C-O} (from carboxylic group)

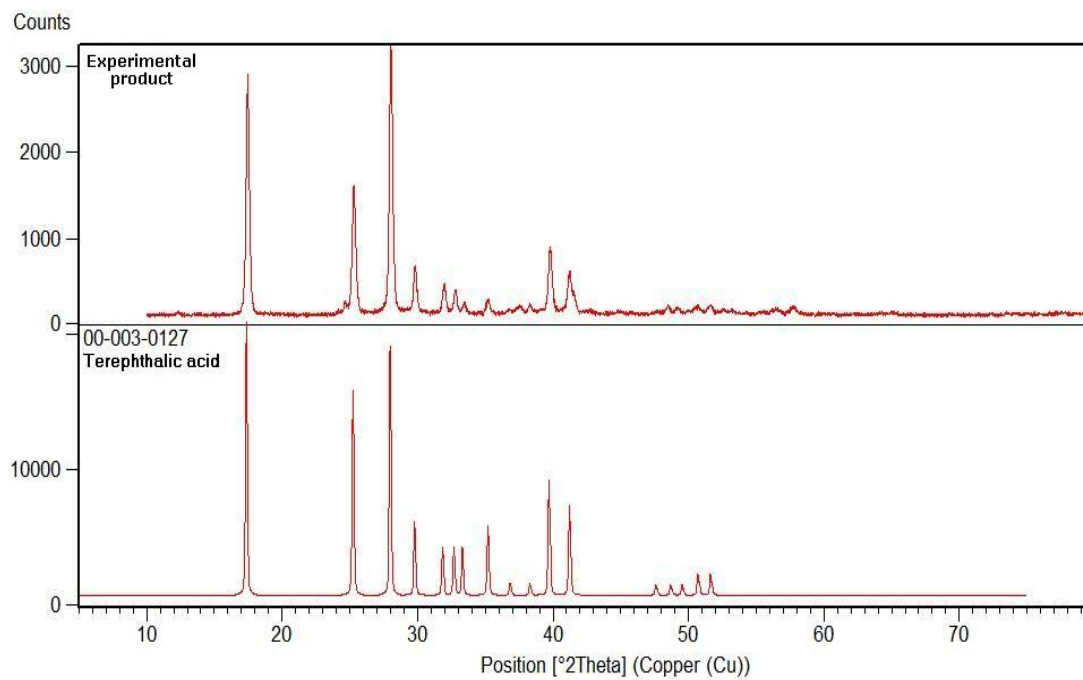


Fig. 3 – X-ray spectrum of TPA powder obtained by alkaline hydrolysis of PET waste indexed to the reference.

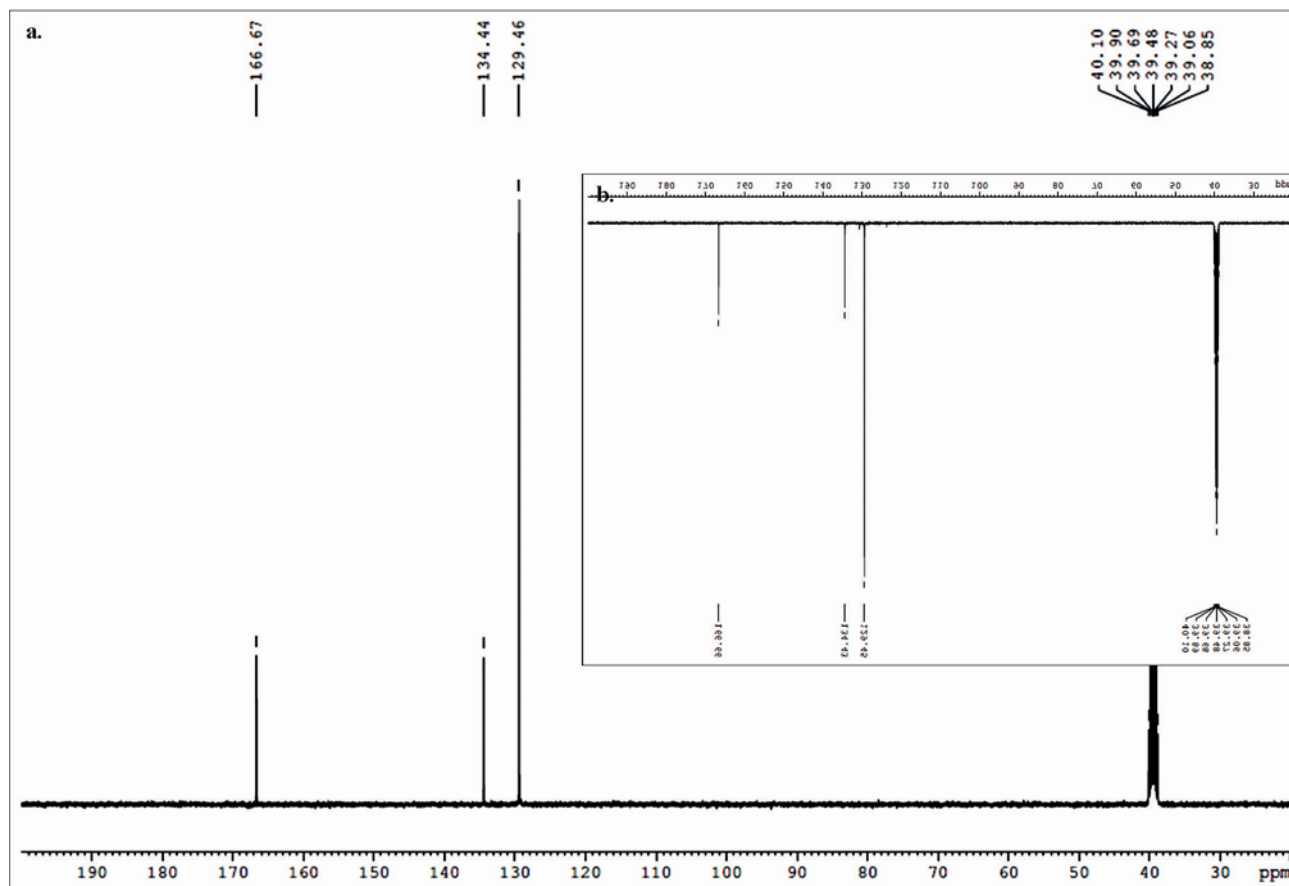


Fig. 4 – ¹³C-NMR spectra of final product (a) and standard TPA (b).

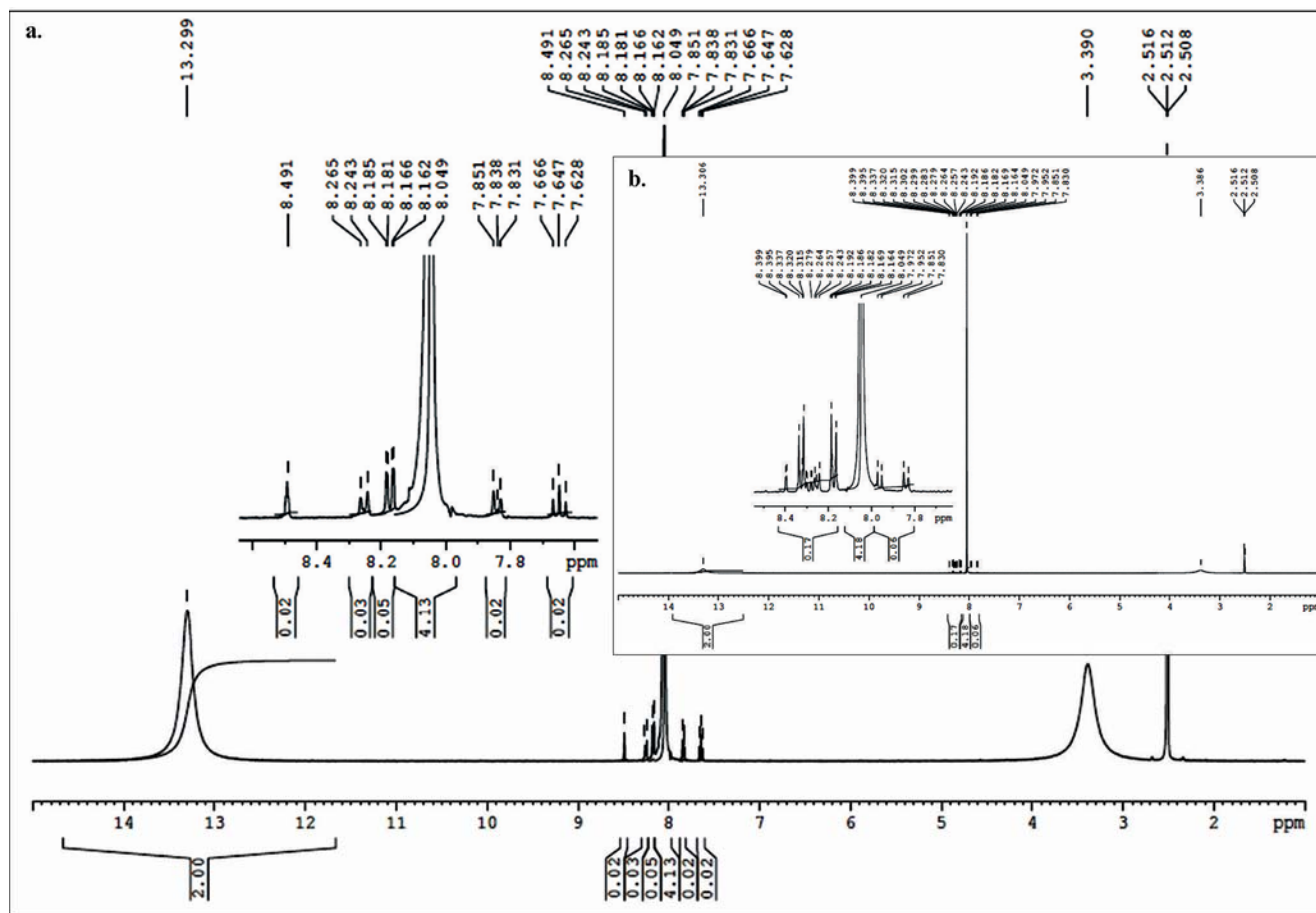


Fig. 5 – ^1H -NMR spectra of final product (a) and standard TPA (b).

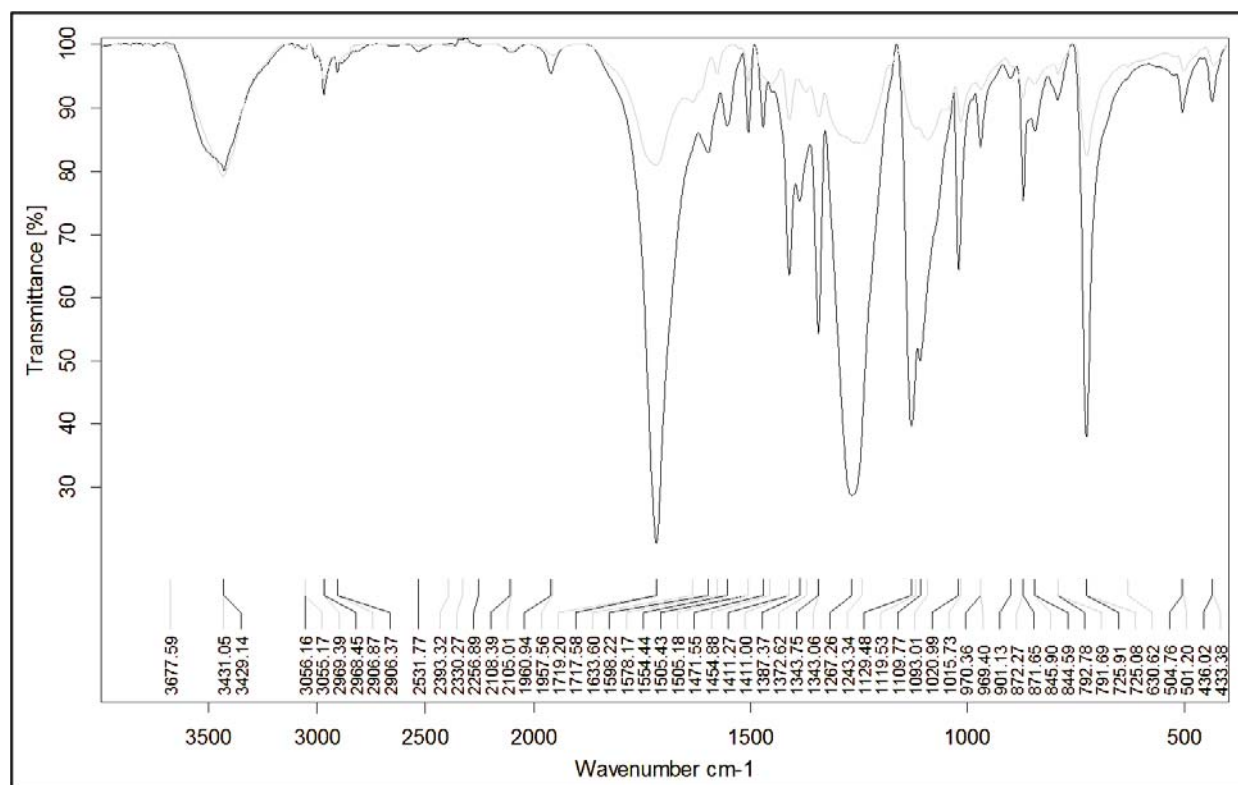


Fig. 6 – FT-IR spectrum of PET waste – raw material (-) and unreacted PET after microwave irradiation (-).

Table 3
Characteristic vibrations of PET

Vibrations	Assignment
$\sim 3430\text{ cm}^{-1}$	$\nu_{\text{O-H}}$
$\sim 2960\text{ cm}^{-1}$	$\nu_{\text{sim}}(\text{CH}_2)$
$\sim 2906\text{ cm}^{-1}$	$\nu_{\text{asim}}(\text{CH}_2)$
$\sim 1719\text{ cm}^{-1}$	$\nu_{\text{C=O}}$
$\sim 1505\text{ cm}^{-1}$	$\delta_{\text{C-H}}$ (deformation in plan)
$\sim 1340\text{ cm}^{-1}$	δ_{CH_2}
$\sim 1240, 1260\text{ cm}^{-1}$	$\nu_{\text{C-O}}$ (from ester group)
$\sim 790, 870\text{ cm}^{-1}$	1,4 substitution of benzene ring

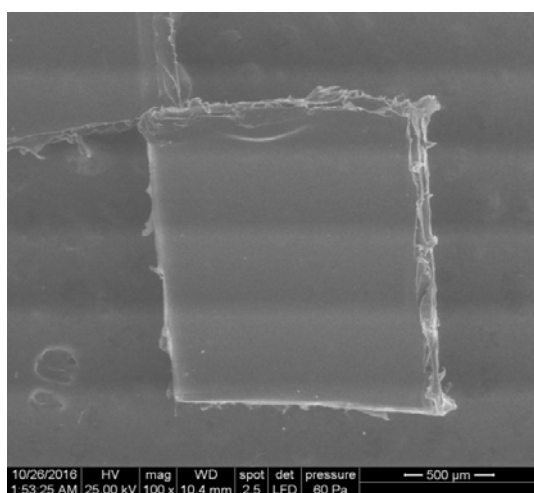


Fig. 7 – SEM image of PET before microwave treatment.

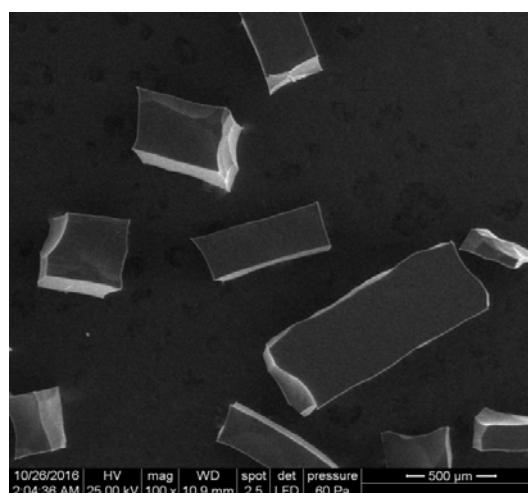


Fig. 8 – SEM image of unreacted PET after microwave treatment.

The ^{13}C -NMR and ^1H -NMR for purified final product vs. standard TPA are shown in Fig. 4 and Fig. 5. The ^{13}C -NMR spectra for sample and standard TPA are identical: $\delta 129.4$ (unsubstituted C of the aromatic ring), 134.4 (C of the aromatic ring to where are attached the carboxyl groups), 166.6 (COOH). The ^1H -NMR spectra proved that the experimental product is terephthalic acid. Besides the signals corresponding to terephthalic acid: $\delta 8.05$ (s, 4H, aromatic H), 13.3 (s, 2H, COOH), in spectra also appear shifts in the range 7.6 - 8.5 , indicating the presence of some aromatic impurities both in standard and experimental products ($<10\%$).

The unreacted PET after completion of hydrolysis reaction was analyzed by recording of FT-IR spectra and scanning electron microscopy (SEM) and the results were compared with those obtained for raw material. The two FT-IR spectra have shown the same characteristic bands (Fig. 6,

Table 3) but with different intensities. Most polymers contain crystalline and amorphous regions; the differences between the bands intensity from IR spectrum can be assigned to different distributions of these regions.

The SEM images of unreacted PET (Fig. 8) compared with raw material (Fig. 7) shows only a reduction of PET flakes dimensions after completion of hydrolysis reaction in microwave field.

CONCLUSIONS

PET hydrolysis in alkaline solution under microwave irradiation was investigated as method for chemical recycling of PET and recovery of obtain pure terephthalic acid in high yields. The use of microwave offers extremely short reaction time for complete degradation of PET waste as

compared to the conventional heating. PET depolymerization was carried out completely for PET: alkaline solution ratio (w/v) of 1 : 10 after 5 minutes of reaction at 220°C, 20 minutes at 210 and 200°C, and 30 minutes at 190°C. TPA yield reached more than 98%. As expected, the decrease of temperature and reaction times lead to lower PET depolymerization degrees. The recorded FT-IR, X-ray and NMR spectra confirmed that the solid product obtained during alkaline hydrolysis under microwave irradiation was high purity terephthalic acid.

Acknowledgments: This work is part of the project PN 09-34 03 07 / 2009 “Studies regarding hydrolysis processes of polyethylene terephthalate waste”, carried out under NUCLEU Program funded by National Authority for Scientific Research (Roumania).

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